

# Adrian Scholl

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## EDUCATION

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### MSc Data Science — Elite Master Program

10/2023 – 08/2026

[Ludwig-Maximilians-Universität München](#) · [Elite Network of Bavaria](#)

- Thesis: *Evaluating Adaptive Timestep Sampling for the Efficient Training of Flow Matching Models* — Computer Vision & Learning group (CompVis), Prof. Dr. Björn Ommer. Ported reinforcement-learned timestep sampling to conditional optimal-transport flow matching, proposed an A2C and a curvature-aware variant, benchmarked on CIFAR-10.
- Seminar: Differentiable Bayesian Causal Structure Learning
- Talk: Bayesian uncertainty quantification for energy price forecasting — international summer school, Hamburg
- Erasmus: MSc Statistics and Operations Research, Universitat Politècnica de Catalunya, Barcelona

### BSc Industrial Engineering and Management

10/2018 – 11/2022

[Karlsruhe Institute of Technology](#)

- Thesis: *Development of an Automated Conflict Prediction System by State Space ARIMA Methods*, in cooperation with the German Federal Foreign Office
- Majors: machine learning, data mining and applications, applications of AI, econometrics, neural networks and genetic algorithms, financial data science, global optimization
- Teaching: tutor for Econometrics (multiple regression in R), Statistics II (~50 students in two courses) and the computer science preparatory courses (Java)
- Germany Scholarship (Deutschlandstipendium), sponsored by QuantCo

## EXPERIENCE

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### Working Student, Machine Learning

11/2024 – 01/2025

[Scalable Capital](#) — [Data Science Unit, Munich](#)

- Built an automated hyperparameter evaluation framework for retrieval-augmented generation (RAG) in AI chatbot agents
- Implemented HyDE retrieval, advanced chunking and re-ranking techniques, and LLM-as-a-judge evaluation (LlamaIndex)
- Cut RAG evaluation runtime through parallel processing and asynchronous API calls

### Working Student, Data Analytics & Analytics Engineering

10/2023 – 09/2024

[FINN](#) — [Fleet Intelligence Unit, Munich](#)

- Solely responsible for ~400 data warehouse models of the fleet department (dbt, GitHub, BigQuery SQL)
- Developed a cash forecasting model enabling management to improve long-term liquidity planning

### Intern, Data Analytics

04/2023 – 09/2023

[FINN](#) — [Fleet Intelligence Unit, Munich](#)

- Owned the project tracking car financings and predicting vehicle underutilization costs via ML (Python, BigQuery SQL)
- Prepared analyses, visualizations and dashboards supporting management decisions and investor requests, leading to \$110m Series C funding (BigQuery SQL, Looker, LookML)
- Built and maintained data warehouse models for KPI tracking, analyses and automations (dbt, BigQuery SQL)

### Data Science Consultant, self-employed

02/2022 – 09/2022

[Auswärtiges Amt \(German Federal Foreign Office\)](#) — [S 05 PREVIEW Crisis Early Warning, Berlin](#)

- Research on crisis early warning with state-space time series models; improved Auto-ARIMA predictions by 56% (MAE)

**Intern, Data Science**

10/2021 – 01/2022

[Auswärtiges Amt — S 05 PREVIEW Crisis Early Warning, Berlin](#)

- Developed an interactive dashboard enabling strategic analysis of the German embassy network (Tableau)
- Consulted on process digitization and created data schemas for a third-party monitoring project in the Middle East
- Created conflict visualizations (maps, charts, dashboards); presented conflict predictions at internal expert workshops

**Intern, Pre-Sales**

06/2018 – 09/2018

[SAP SE — Public Services / Smart Cities, Walldorf](#)

- Analyzed the smart city start-up market and evaluated partnership opportunities; consulted on SAP IoT, S/4HANA and AI/ML use cases

SKILLS & PROJECTS

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|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LANGUAGES    | German (native), English (C1), Spanish (B2), French (A2)                                                                                                                                                                                                                                                                                                         |
| AI / ML      | PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, LlamaIndex, statsmodels, sktime, SciPy, CVXPY; flow matching and diffusion, reinforcement learning (REINFORCE, A2C), Bayesian inference, state-space models                                                                                                                                                    |
| SOFTWARE DEV | Python, Java, Git, CI/CD, FastAPI, Pydantic, SQLAlchemy                                                                                                                                                                                                                                                                                                          |
| DATA         | SQL, R, BigQuery, dbt, Looker, LookML, Tableau, GSheets, Excel                                                                                                                                                                                                                                                                                                   |
| PROJECTS     | VISTA — web app tracking attention level while studying via computer vision · EthXPose — identifying fraudulent Ethereum wallets from transaction graph embeddings · Bayesian energy price modelling under arbitrage constraints · Default prediction for commercial real estate mortgages · PISA score prediction with pooled OLS and clustered standard errors |